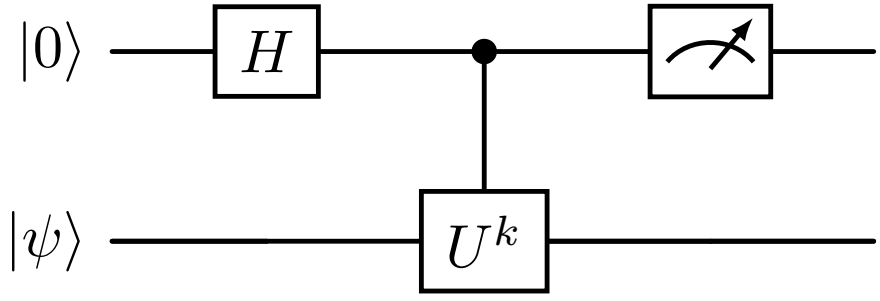


measure the observable X (or Y)

$E[X] = \mathrm{Re}\,\langle\psi|U^k|\psi\rangle, \qquad E[Y] = \mathrm{Im}\,\langle\psi|U^k|\psi\rangle,$ each shot gives $+1$ or -1